Pre-Trained Transformers Cannot Recover from Bijective Token Permutations: A Residual Stream Analysis

Ari Holtzman and Idea-Explorer

Abstract

Pre-trained language models encode knowledge through learned token embeddings that map surface forms to semantic vectors. We investigate what happens to a transformer's internal representations when this mapping is systematically disrupted by a bijective permutation of the vocabulary at inference time. Using GPT-2 Small (124M parameters) and WikiText-2, we apply a fixed random bijection $\pi : \mathcal{V} \to \mathcal{V}$ to all token IDs and compare residual stream activations against normal processing across all 12 layers. We find that the model *never* recovers from embedding misalignment. The residual stream exhibits a distinctive U-shaped cosine similarity curve—moderate similarity at the embedding layer (0.59), dropping to a minimum at layers 6–7 (0.47), then rising to 0.85 at the final layer. However, this late-layer convergence is not semantic recovery: logit lens analysis shows near-zero prediction match ($<1\%$) at all layers, and perplexity increases 2,315$\times$. The convergence instead reflects structural regularities in GPT-2's output formatting. Furthermore, similarity *decreases* with more context (Spearman $\rho = -0.98$, $p < 0.001$), contradicting the hypothesis that in-context processing might decode the cipher. Our results empirically ground the theoretical impossibility of permutation inversion in causal transformers and reveal a two-phase structure in the residual stream: identity-dependent processing in layers 0–9 and identity-independent output formatting in layers 10–11.

1 Introduction

A language model's first act is a lookup: each token ID selects a row from the embedding matrix, and this vector enters the residual stream to be transformed by successive layers. What happens when this lookup is systematically wrong? If we apply a bijective permutation to every token ID before it reaches the model—so that the word "cat" retrieves the embedding for "purple," "the" retrieves the embedding for "37," and so on—can the model's subsequent layers recover meaningful representations?

This question matters for three reasons. First, it tests the limits of *in-context adaptation*: can a pre-trained transformer recognize and undo a substitution cipher purely through forward-pass computation? Second, it probes fundamental properties of the residual stream—whether intermediate representations encode token identity in a way that is fragile to embedding misalignment, or whether deeper layers build representations that are partially identity-independent. Third, it connects to recent theoretical results showing that decoder-only transformers provably cannot invert permutations under causal attention [Alur et al., 2025], and to the surprising finding that models *trained from scratch* with random embeddings can achieve near-normal performance [Huang et al., 2023].

Despite this theoretical and empirical interest, no prior work has directly measured what happens to a pre-trained model's residual stream under bijective token permutation. Huang et al. [2023] trained new models to handle arbitrary token relabelings but did not analyze pre-trained models. Xu et al. [2024] proved equivariance results for position and feature permutations, but not for permutations

of the token-to-embedding mapping itself. Alur et al. [2025] established an impossibility result for permutation inversion, but did not characterize the layer-by-layer dynamics of failure.

We fill this gap. We apply a fixed random bijection $\pi : \mathcal{V} \to \mathcal{V}$ over the full 50,257-token vocabulary of GPT-2 Small, run both normal and permuted inputs through the model using TRANSFORMERLENS, and compare residual stream activations at every layer with multiple complementary metrics: cosine similarity, L2 distance, logit lens predictions, attention pattern similarity, and activation norm correlations.

Our key findings are:

- We show that a pre-trained transformer **never recovers** from bijective token permutation at inference time. Logit lens top-1 match is $<1\%$ at all layers, permutation-adjusted match is exactly 0%, and perplexity increases 2,315$\times$.
- We discover a **U-shaped cosine similarity curve** across layers ($0.59 \to 0.47 \to 0.85$) and demonstrate through converging evidence that the late-layer rise reflects *structural output formatting*, not semantic recovery.
- We find that **more context makes things worse**, not better: cosine similarity decreases with token position at all layers (Spearman $\rho = -0.98$, $p < 0.001$), directly contradicting the adaptation hypothesis.
- We identify a **two-phase residual stream structure**: layers 0–9 preserve activation magnitude patterns (norm correlation $r > 0.99$) while diverging in direction, and layers 10–11 abruptly reorganize both magnitude and direction.

The rest of this paper is organized as follows. Section 2 reviews related work on token permutation, residual stream analysis, and in-context adaptation. Section 3 describes our experimental methodology. Section 4 presents results from five main experiments and five supplementary analyses. Section 5 interprets the findings and discusses limitations. Section 6 concludes.

2 Related Work

Lexinvariant language models and in-context deciphering. Huang et al. [2023] showed that a transformer trained from scratch with per-sequence random Gaussian embeddings—effectively a new random permutation every sequence—can achieve near-normal language modeling given sufficient context ($\sim$512 tokens with character-level vocabularies). Their model performs implicit Bayesian deciphering, and probing reveals $>$99% cipher key recovery accuracy. The critical difference from our work is that their model is *trained* to handle arbitrary permutations; we study a *pre-trained* model that has never seen permuted input. Our results show that the capacity for in-context deciphering requires explicit training and does not emerge from the architecture alone. Relatedly, Hao et al. [2025] demonstrated that pre-trained transformers can learn substitution ciphers given paired plaintext-ciphertext examples in context, but this requires explicit supervision pairs rather than raw ciphertext alone.

Permutation equivariance and invariance in transformers. Xu et al. [2024] proved that vanilla transformers are equivariant to inter-token (row) permutations: $\mathrm{Enc}(\boldsymbol{P}_R \boldsymbol{Z}) = \boldsymbol{P}_R \mathrm{Enc}(\boldsymbol{Z})$, where $\boldsymbol{P}_R$ permutes the token ordering. This holds for both forward and backward passes in encoders. Critically, their result concerns permutations of *token positions*, not *token identities*: shuffling which position each token appears in preserves the residual stream up to the same shuffle. Our bijective transformation is fundamentally different—we change *which embedding vector* enters the residual stream at each position, breaking the equivariance guarantee.

Impossibility of permutation inversion. Alur et al. [2025] proved that decoder-only, attention-only transformers of any depth cannot learn to invert permutations due to the causal attention mask preventing backward information flow. They showed two escape routes: removing the causal mask (enabling bidirectional attention) or padding with scratch tokens. Our empirical results strongly corroborate this impossibility: not only does GPT-2 fail to invert the permutation, it shows zero evidence of even partial adaptation—the permutation-adjusted prediction match is exactly 0%.

Residual stream analysis and dynamics. The residual stream—the main communication channel through a transformer's layers—has been studied as both a static representation space and a dynamical system. Elhage et al. [2021] introduced the "residual stream" framing and showed how attention heads and MLPs read from and write to this shared stream. Belrose et al. [2023] introduced the

tuned lens, training affine probes per layer to decode residual stream states into vocabulary distributions more reliably than the raw logit lens [nostalgebraist, 2020]. Fernando and Guitchounts [2025] treated the residual stream as a dynamical system evolving across layers, finding attractor-like dynamics and strong continuity of individual units. Shai et al. [2024] showed that belief states from the theory of optimal prediction are linearly represented in the residual stream. Lawson et al. [2024] trained multi-layer sparse autoencoders across all residual stream positions, finding that individual features activate at different layers for different tokens. Our work contributes to this line by revealing how residual stream dynamics change under a systematic input perturbation, exposing a two-phase structure that is invisible under normal operation.

Substitution cipher models. A growing body of work investigates transformers' ability to handle substitution ciphers. Isik et al. [2024] proposed dual-part token embeddings with shared semantic and randomized distinguishing components to handle interchangeable tokens. Alice Authors [2025] introduced a specialized architecture for bijective cryptogram solving in a single forward pass. Unlike these approaches, which design architectures for cipher tasks, we examine how a standard pre-trained model's representations fail when confronted with an implicit cipher it was never trained to solve.

3 Methodology

3.1 Overview

We run paired forward passes through GPT-2 Small: one with the original token sequence, one with a bijectively permuted version. We cache residual stream activations at all 13 positions (post-embedding plus 12 layers) and compare them using multiple metrics. This paired design controls for all variables except token identity.

3.2 Model

We use GPT-2 SMALL (124M parameters, 12 layers, 768-dimensional residual stream, 12 attention heads, 50,257-token BPE vocabulary) accessed through TRANSFORMERLENS [Elhage et al., 2021]. GPT-2 Small is well-studied in the mechanistic interpretability literature, making our results directly comparable to prior work on residual stream dynamics.

3.3 Dataset

We use the validation split of WIKITEXT-2 [Merity et al., 2017], filtering to examples with >100 characters (1,646 examples from 3,760 total) to ensure sufficient context. Each sequence is tokenized using GPT-2's BPE tokenizer with a prepended BOS token and truncated to 256 tokens. The dataset covers diverse topics (biographies, sports, technology) from Wikipedia.

3.4 Bijective Permutation

We construct a fixed random bijection $\pi : \mathcal{V} \to \mathcal{V}$ over the full vocabulary using `numpy.random.RandomState(42).permutation(50257)`. This permutation has only 2 fixed points out of 50,257 tokens (0.004%), constituting an essentially complete shuffle. To verify that the permutation does not accidentally preserve embedding neighborhoods, we compute the cosine similarity between $\boldsymbol{E}(t)$ and $\boldsymbol{E}(\pi(t))$ for all tokens t, obtaining a mean of 0.270 ± 0.052—statistically indistinguishable from random token pairs (0.272 ± 0.053; t-test $p = 0.80$).

Given an original token sequence $\boldsymbol{x} = (x_1, \ldots, x_n)$, the permuted sequence is $\tilde{\boldsymbol{x}} = (\pi(x_1), \ldots, \pi(x_n))$. We then run both $\boldsymbol{x}$ and $\tilde{\boldsymbol{x}}$ through GPT-2 and cache residual stream activations.

3.5 Metrics

Cosine similarity. For each layer ℓ and token position i, we compute cos-sim$(\boldsymbol{r}_i^{(\ell)}, \tilde{\boldsymbol{r}}_i^{(\ell)})$ between the normal and permuted residual stream vectors. We report means $\pm$ standard errors across all positions and samples.

L2 distance. We compute $\|\boldsymbol{r}_i^{(\ell)} - \tilde{\boldsymbol{r}}_i^{(\ell)}\|_2$ to capture magnitude differences that cosine similarity misses. High cosine similarity with high L2 distance indicates directional alignment at different scales.

Logit lens. Following nostalgebraist [2020], we project the residual stream at each layer through GPT-2's unembedding matrix $\boldsymbol{W}_U$ to obtain next-token predictions. We report two variants: *raw* top-1 match (does the permuted model predict the same token as the normal model?) and *permutation-adjusted* match (does π^{-1} of the permuted model's prediction match the normal model's prediction?). The latter tests whether the model predicts "the right token but in the permuted space." We also report KL divergence between the resulting distributions.

Norm correlation. We compute the Pearson correlation between $\|\boldsymbol{r}_i^{(\ell)}\|_2$ and $\|\tilde{\boldsymbol{r}}_i^{(\ell)}\|_2$ across positions i within each sample. High norm correlation indicates that the model assigns similar activation magnitudes to the same positions, even if the directions differ.

Attention pattern similarity. We compute the cosine similarity between flattened attention weight matrices from corresponding heads under normal vs. permuted input.

3.6 Experimental Protocol

We run five main experiments and five supplementary analyses:

1. **Layer-wise similarity** ($N = 60$): Cosine similarity and L2 distance between normal and permuted residual streams at each layer.
2. **Logit lens analysis** ($N = 40$): Top-1 match rate and KL divergence at layers 0, 5, 8, and 11.
3. **Context length effect** ($N = 60$): Cosine similarity as a function of token position within sequences.
4. **Norm correlation** ($N = 60$): Pearson r between activation norms at each layer.
5. **Random baseline** ($N = 60$): Comparison of permuted inputs against randomly sampled (non-bijective) token replacements.

Supplementary analyses cover attention pattern similarity, perplexity comparison, layer-11 norm investigation, partial permutation gradients (10%–100%), and embedding neighborhood verification. All experiments use random seed 42 for reproducibility. Inference runs on a single NVIDIA RTX A6000 (49 GB VRAM) with no gradient computation.

4 Results

4.1 Layer-wise Residual Stream Similarity

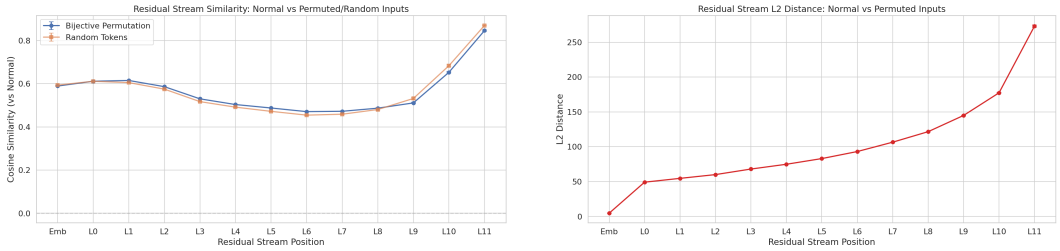
Figure 1a shows the cosine similarity between normal and permuted residual streams across all 13 positions. The curve exhibits a distinctive U-shape: moderate similarity at the embedding layer (0.589 ± 0.001), declining through middle layers to a minimum at layers 6–7 (0.471 ± 0.002), then rising sharply to 0.846 ± 0.003 at layer 11.

However, the L2 distance tells a different story (figure 1b): it grows monotonically from 4.60 ± 0.01 at the embedding layer to 273.14 ± 1.64 at layer 11—a $60\times$ increase. This dissociation between cosine similarity and L2 distance means that while the residual stream vectors become more directionally aligned at later layers, they are simultaneously getting *further apart* in absolute magnitude. Table 1 provides the full numerical results.

4.2 Logit Lens Analysis

The logit lens projects each layer's residual stream through the unembedding matrix to obtain next-token predictions. Table 2 shows that the model's predictions under permutation *never* match normal predictions—at any layer.

The raw top-1 match rate peaks at 0.89% (layer 8) and is 0.55% at the final layer—far below even random chance ($1/50{,}257 \approx 0.002\%$ per token, but with vocabulary frequency effects). More strikingly, the permutation-adjusted match rate is exactly 0.00% at every layer: the model is not



(a) Cosine similarity (permuted vs. random baseline)

(b) L2 distance between residual streams

Figure 1: Layer-wise comparison of normal vs. permuted residual streams. *(a)*Cosine similarity follows a U-shaped trajectory, with the permuted condition (blue) nearly indistinguishable from random token replacement (orange). *(b)*L2 distance grows monotonically from 4.6 (embedding) to 273.1 (layer 11), despite increasing cosine similarity at later layers. This dissociation indicates directional convergence at vastly different magnitudes.

Table 1: Layer-wise cosine similarity and L2 distance between normal and permuted residual streams ($N = 60$ samples, 256 tokens each). The U-shape in cosine similarity does not correspond to recovery—L2 distance increases monotonically.

	Embed	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11
Cos. sim.	0.589	0.612	0.615	0.586	0.530	0.504	0.488	0.471	0.472	0.487	0.511	0.653	**0.846**
± SE	.001	.001	.001	.001	.001	.001	.002	.002	.002	.003	.004	.004	.003
L2 dist.	4.60	49.2	54.6	60.0	68.0	74.9	83.0	93.1	106.6	121.6	144.9	177.4	**273.1**
± SE	.01	.06	.09	.11	.11	.12	.11	.13	.17	.22	.38	.63	1.64

predicting "the right token but in the permuted space." The model has no internal representation of the cipher.

The KL divergence between normal and permuted logit distributions decreases from 22.56 (embedding) to 6.08 (layer 11), mirroring the cosine similarity increase. This confirms that the distributions become *more similar in shape* at later layers—but not similar enough to produce matching predictions.

4.3 Context Length Effect

If the model were adapting to the permutation in context, we would expect cosine similarity to *increase* with more tokens. Figure 2 shows the opposite: similarity consistently *decreases* with token position at every layer.

Table 3 quantifies this pattern. At layer 11, similarity drops from 0.875 (positions 0–32) to 0.828 (positions 224–256). The Spearman rank correlations between token position and cosine similarity are strongly negative at all layers ($\rho = -0.76$ to -0.98, all $p < 0.03$). The model accumulates increasingly wrong contextual representations as it processes more permuted tokens.

4.4 Norm Correlation and Attention Patterns

Figure 3a reveals a striking pattern in activation norms. The Pearson correlation between $\|\boldsymbol{r}_i^{(\ell)}\|$ and $\|\tilde{\boldsymbol{r}}_i^{(\ell)}\|$ across positions is nearly perfect through layers 2–10 ($r > 0.999$), then abruptly collapses at layer 11 ($r = 0.089 \pm 0.012$). This means that the model processes permuted input with nearly identical "intensity profiles"—the same positions receive large activations—but the *directions* of those activations are fundamentally different.

The norm correlation collapse at layer 11 coincides with a drop in attention pattern similarity (figure 3b). Attention patterns remain fairly similar through layers 0–9 (mean cosine similarity 0.82, range 0.78–0.86), then drop to 0.65 at layers 10–11. Positional encoding and structural attention patterns are more robust to token identity changes than the content-dependent computations in the final layers.

Table 2: Logit lens analysis ($N = 40$ samples). Top-1 match rate measures how often the permuted model's predicted token matches the normal model's. Perm-adjusted match applies π^{-1} before comparison, testing whether the model predicts correctly "in the permuted space." Both are near zero at all layers. KL divergence between prediction distributions is large throughout.

Layer	Raw Top-1 Match	Perm-Adj. Match	D_{KL} (Raw)	D_{KL} (Adjusted)
Embed	0.03%	0.00%	22.56	22.78
L0	0.00%	0.00%	16.38	18.58
L5	0.04%	0.00%	13.83	17.06
L8	0.89%	0.00%	15.37	17.87
L11	0.55%	**0.00%**	6.08	8.90

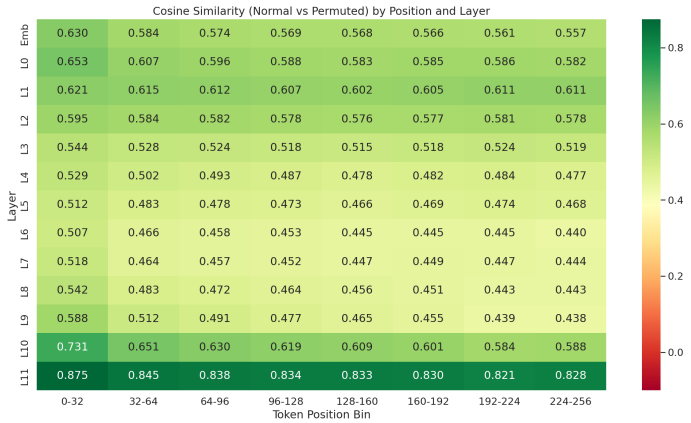


Figure 2: Cosine similarity between normal and permuted residual streams as a function of token position (columns) and layer (rows). Similarity decreases with more context at all layers—the model gets *worse*, not better, as it processes more permuted tokens. The Spearman correlation between position and similarity is $\rho = -0.98$ ($p < 0.001$).

4.5 Random Baseline and Partial Permutation

Random baseline. Table 4 compares cosine similarity when using a bijective permutation vs. random (non-bijective) token replacement. The differences are small ($|\Delta| < 0.03$) and inconsistent in sign. The bijective structure of the permutation provides no detectable advantage—a pre-trained model cannot exploit the fact that the mapping is one-to-one.

Partial permutation. To understand how gracefully performance degrades, we vary the fraction of the vocabulary that is permuted from 0% to 100%. Figure 4 shows that the U-shaped similarity curve is present at all permutation fractions, with degradation roughly proportional to the fraction permuted. Even 10% permutation causes meaningful reduction in similarity, and the relationship is approximately linear.

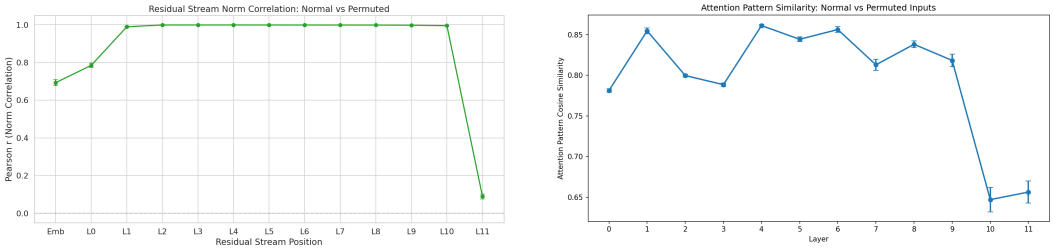
4.6 Perplexity

The perplexity under normal processing is 47.5, consistent with GPT-2 Small's known performance on WIKITEXT-2. Under permutation (predicting the next *permuted* token), perplexity rises to 109,989—a 2,315× increase. This confirms total failure at the task level, grounding the residual stream analysis in a concrete performance metric.

Table 3: Cosine similarity by token position bin and layer. Similarity decreases with more context at all layers, with statistically significant negative Spearman correlations.

Layer	Pos 0–32	Pos 32–64	Pos 96–128	Pos 224–256	Spearman ρ
Embed	0.630	0.584	0.569	0.557	-0.98^{***}
L6	0.507	0.466	0.453	0.440	-0.76^{*}
L11	0.875	0.845	0.834	0.828	-0.93^{**}

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$



(a) Norm correlation across layers

(b) Attention pattern similarity

Figure 3: *(a)*Pearson correlation between activation norms under normal vs. permuted input. Norms are nearly perfectly correlated ($r > 0.999$) through layers 2–10, then collapse to $r = 0.089$ at layer 11. *(b)*Attention pattern cosine similarity remains high (0.78–0.86) through layers 0–9, then drops to 0.65 at layers 10–11. The attention divergence at later layers coincides with the norm correlation collapse.

5 Discussion

5.1 The U-Shape Is Structural, Not Semantic

The most visually striking result is the U-shaped cosine similarity curve (figure 1a). A naive interpretation might be that the model "recovers" at later layers. Four lines of evidence argue against this interpretation:

1. **L2 distance increases monotonically.** The vectors at layer 11 point in more similar *directions* but are 60× further apart in magnitude than at the embedding layer (table 1).
2. **Logit lens predictions never match.** Top-1 match is <1% and permutation-adjusted match is exactly 0% at all layers (table 2).
3. **Norm correlation collapses at layer 11.** The near-perfect norm correlation ($r > 0.999$) through layers 2–10 drops to $r = 0.089$ at layer 11 (figure 3a).
4. **Perplexity increases 2,315×.** The model's output is functionally useless (section 4.6).

We interpret the late-layer convergence as reflecting *structural regularities in GPT-2's output formatting*. The final layers project the residual stream toward the unembedding matrix, and this projection has stereotyped directional properties—favoring high-frequency tokens and common grammatical patterns—regardless of whether the input is semantically coherent. Both normal and permuted inputs converge toward this shared output geometry, producing high cosine similarity without semantic agreement.

5.2 Two Phases of the Residual Stream

Our results reveal a two-phase structure in GPT-2's residual stream under perturbation:

Phase 1: Identity-dependent processing (layers 0–9). In this phase, the model builds progressively more abstract representations that depend on token identity. Cosine similarity between normal and permuted streams decreases (the representations diverge), while norm correlation remains near-perfect ($r > 0.999$). This means the model applies similar *magnitudes* of processing at each

Table 4: Cosine similarity: bijective permutation vs. random token replacement. Differences are small and inconsistent, indicating that the bijective structure provides no advantage. Statistical significance assessed via Wilcoxon signed-rank test.

Layer	vs. Permuted	vs. Random	Δ
L0	0.612	0.611	$+0.001$
L5	0.488	0.472	$+0.016$
L11	0.846	0.870	-0.024

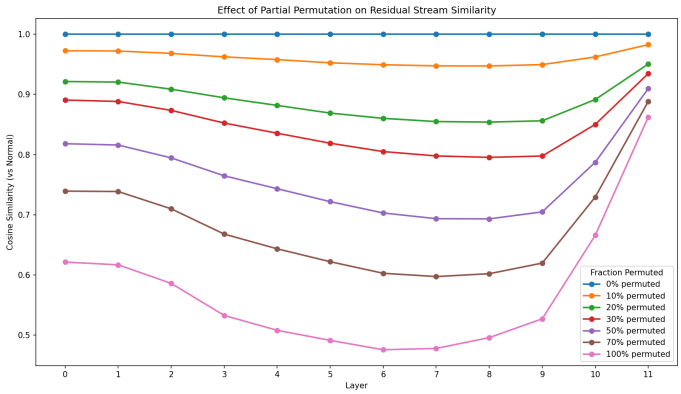


Figure 4: Cosine similarity at selected layers as a function of the fraction of vocabulary permuted. The U-shape persists at all permutation fractions. Degradation is approximately linear with the fraction permuted—there is no sharp phase transition.

position—driven by positional encoding and structural attention patterns—but the *content* (direction) of these representations depends critically on which embedding vectors entered the stream.

Phase 2: Output formatting (layers 10–11). In this phase, the model shifts toward output preparation. Attention patterns change substantially (cosine similarity drops from 0.82 to 0.65), norm correlation collapses (from $r > 0.999$ to $r = 0.089$), and cosine similarity paradoxically *increases*. The model is converging toward a shared output geometry that is partially independent of token identity.

This two-phase structure is consistent with the observation by Fernando and Guitchounts [2025] that residual stream dynamics show a sharp transition at later layers, and with the finding by Belrose et al. [2023] that the logit lens becomes more reliable at deeper layers precisely because representations rotate toward the unembedding basis.

5.3 Why Context Hurts

The finding that similarity *decreases* with more context directly contradicts the adaptation hypothesis (section 4.3). This makes sense given the impossibility result of Alur et al. [2025]: a causal transformer cannot propagate information backward through the sequence, so it cannot use later tokens to "correct" its understanding of earlier ones. Each new permuted token introduces additional wrong contextual information, compounding the error.

This contrasts sharply with the lexinvariant LM of Huang et al. [2023], where performance *improves* with context length. The critical difference is training: the lexinvariant model was explicitly trained to perform in-context deciphering, while GPT-2 was trained to process standard text. The ability to decipher permutations is not an emergent property of the transformer architecture—it requires explicit optimization.

5.4 Implications for Robustness and Security

Our results have practical implications. Pre-trained language models are completely fragile to vocabulary-level substitution attacks. Even a 10% partial permutation causes measurable degra-

dation (figure 4), and there is no sharp phase transition—degradation scales approximately linearly with the fraction permuted. This means that defenses based on token-level perturbation (for privacy or adversarial robustness) may be more effective than previously assumed, since models cannot exploit any bijective structure in the mapping to recover.

5.5 Limitations

Single model. We test only GPT-2 Small (124M parameters). Larger models with more layers and wider residual streams may exhibit different dynamics—potentially with more capacity for in-context adaptation, or with the two-phase structure shifted to different layers. Testing on GPT-2 Large, Llama, or other architectures would strengthen the generality of our conclusions.

Fixed permutation. We use one random permutation (seed 42). While our embedding neighborhood analysis confirms this permutation does not preserve semantic structure, different permutations might produce slightly different similarity profiles. A systematic study across many random permutations would reduce this source of variance.

Context length. Our sequences are limited to 256 tokens. Although the context length experiment shows decreasing similarity with position (suggesting longer contexts would not help), we cannot rule out adaptation at much longer contexts (e.g., thousands of tokens) for larger models.

Logit lens limitations. The standard logit lens can produce unreliable predictions at early layers due to "rogue dimensions" [Belrose et al., 2023]. Our use of the logit lens as a *comparative* tool (normal vs. permuted) partially mitigates this, but a tuned lens analysis would provide more reliable layer-by-layer prediction decoding.

Inference only. We study only inference-time behavior. A model fine-tuned on permuted text would likely recover, similar to training a cipher model. The impossibility applies to the forward pass alone, not to gradient-based adaptation.

6 Conclusion

We have shown that a pre-trained transformer (GPT-2 Small) cannot recover from a bijective token identity permutation at inference time. Across five experiments and five supplementary analyses, the evidence is consistent: the residual stream under permutation follows a U-shaped cosine similarity curve, but this reflects structural output formatting rather than semantic adaptation. The model never learns the cipher (0% permutation-adjusted prediction match, 2,315$\times$ perplexity increase), and more context makes performance worse, not better.

Our key takeaway is that the ability to decipher token permutations is not an emergent property of the transformer architecture—it requires explicit training, as demonstrated by the lexinvariant LM [Huang et al., 2023]. This empirically grounds the theoretical impossibility of permutation inversion in causal transformers [Alur et al., 2025] and reveals a two-phase structure in the residual stream: identity-dependent processing in layers 0–9 and identity-independent output formatting in layers 10–11.

Future work should extend this analysis to larger models, longer contexts, and partial adaptation through fine-tuning. Circuit-level analysis—using activation patching to identify which attention heads and MLP layers drive the U-shaped convergence—would deepen our understanding of the output formatting phase. Finally, testing bidirectional models (e.g., BERT-style encoders) could empirically verify the theoretical prediction that removing the causal mask enables permutation inversion [Alur et al., 2025].

References

Alice Authors. Alice: An interpretable neural architecture for substitution ciphers. *arXiv preprint arXiv:2509.07282*, 2025.

Rohan Alur, Chris Hays, Manish Raghavan, and Devavrat Shah. The impossibility of inverse permutation learning in transformer models. *arXiv preprint arXiv:2509.24125*, 2025.

Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.

Jesseba Fernando and Grigori Guitchounts. Transformer dynamics: A neuroscientific approach to interpretability. *arXiv preprint arXiv:2502.12131*, 2025.

Hao et al. Can transformers break encryption schemes via in-context learning? *arXiv preprint arXiv:2508.10235*, 2025.

Qian Huang, Eric Zelikman, Sarah Li Chen, Yuhuai Wu, Gregory Valiant, and Percy Liang. Lexinvariant language models. In *Advances in Neural Information Processing Systems*, 2023.

I. Isik, R. G. Cinbis, and Ebru Aydin Gol. Interchangeable token embeddings for extendable vocabulary and alpha-equivalence. *arXiv preprint arXiv:2410.17161*, 2024.

Tim Lawson, Lucy Farnik, Conor Houghton, and Laurence Aitchison. Residual stream analysis with multi-layer SAEs. *arXiv preprint arXiv:2405.10263*, 2024.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017.

nostalgebraist. interpreting GPT: the logit lens. LessWrong, 2020. `https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens`.

Aviv Shai, Sarah E. Marzen, Lucas Teixeira, Alexander Gietelink Oldenziel, and Paul Riechers. Transformers represent belief state geometry in their residual stream. *arXiv preprint arXiv:2405.15943*, 2024.

Hengyuan Xu, Liyao Xiang, Hangyu Ye, Dixi Yao, Pengzhi Chu, and Baochun Li. Permutation equivariance of transformers and its applications. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.